

Please make sure that you show all of the steps, not just the answer.

1) $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = \frac{x + 5}{(5) + 5} = \boxed{10}$$

$$\frac{x^2 - 5^2}{x - 5} = \frac{(x - 5)(x + 5)}{(x - 5)}$$

3) $\lim_{x \rightarrow 4} \frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$

Combine fractions in the numerator.
Common denominator is $4x$

$$\begin{aligned} \frac{\frac{4-x}{4x}}{x-4} &= \frac{4-x}{4x} \cdot \frac{1}{x-4} \\ &= \frac{-(x-4)}{4x} \cdot \frac{1}{x-4} \\ &= \frac{-1}{4x} = \lim_{x \rightarrow 4} -\frac{1}{4(4)} = \boxed{-\frac{1}{16}} \end{aligned}$$

5) $\lim_{x \rightarrow 5} \frac{x^3 - 125}{x^2 - 25}$

$\frac{x^3 - 5^3}{x^2 - 5^2} \rightarrow \frac{\text{Difference of cubes}}{\text{Difference of squares}}$

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{(x-5)(x^2 + 5x + 25)}{(x-5)(x+5)} &= \frac{(5)^2 + (5)5 + 25}{(5) + 5} \\ &= \frac{75}{10} = \boxed{\frac{15}{2}} \end{aligned}$$

2) $\lim_{x \rightarrow -2} \frac{x^2 - 5x}{x^3 + 2}$

$$\frac{(-2)^2 - 5(-2)}{(-2)^3 + 2}$$

$$\frac{4 + 10}{-8 + 2}$$

$$\frac{14}{-6} = \boxed{-\frac{7}{3}}$$

4) $\lim_{h \rightarrow 0} \frac{-(x+h)^3 + 2(x+h) + x^3 - 2x}{h}$

$$\begin{aligned} -(x+h)^3 &= -(x^3 + 3x^2h + 3xh^2 + h^3) \\ &= -x^3 - 3x^2h - 3xh^2 - h^3 \end{aligned}$$

$$2(x+h) = 2x + 2h$$

$$\begin{aligned} &= -x^3 - 3x^2h - 3xh^2 - h^3 + 2x + 2h \\ &= \dots + x^3 - 2x \end{aligned}$$

$$\begin{aligned} &= \frac{h(-3x^2 - 3xh - h^2 + 2)}{h} \quad (\text{typo}) \\ &= -3x^2 - 3xh - h^2 + 2 \end{aligned}$$

$$\begin{aligned} h \rightarrow 0 &= -3x^2 - 3x(0) - (0)^2 + 2 \\ &= -3x^2 + 2 \end{aligned}$$

6) $\lim_{x \rightarrow \infty} \frac{12x^2 - 7x}{6x^2 - 5x}$

Since $x \rightarrow \infty$, identify highest power (x^2) of denominator

$$\begin{aligned} \frac{\frac{12x^2}{x^2} - \frac{7x}{x^2}}{\frac{6x^2}{x^2} - \frac{5x}{x^2}} &= \frac{12 - \frac{7}{x}}{6 - \frac{5}{x}} \quad \left. \begin{array}{l} \text{as } x \rightarrow \infty \\ \frac{7}{x} \rightarrow 0 \\ \frac{5}{x} \rightarrow 0 \end{array} \right\} \\ &= \frac{12 - 0}{6 - 0} = \frac{12}{6} = \boxed{2} \end{aligned}$$

$$7) \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$\frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$\frac{h(3x^2 + 3xh + h^2)}{h}$$

$$\lim_{h \rightarrow 0} 3x^2 + 3x(0) + (0)^2 = \boxed{3x^2}$$

$$8) \lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25} \cdot \frac{\sqrt{x} + 5}{\sqrt{x} + 5}$$

$$\frac{x - 25}{(x - 25)(\sqrt{x} + 5)}$$

$$\frac{1}{\sqrt{x} + 5} \quad \lim_{x \rightarrow 25} \frac{1}{\sqrt{(25)} + 5}$$

$$= \frac{1}{5 + 5} = \boxed{\frac{1}{10}}$$

9) Given the function $f(x) = \frac{1}{x^2}$, evaluate $\lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4}$.

$$\frac{\frac{1}{x^2} - \frac{1}{16}}{x - 4}$$

common factor is $16x^2$

$$\frac{16 - x^2}{16x^2} = \frac{(16 - x^2)}{16x^2} \cdot \frac{1}{x - 4}$$

$$\frac{16 - x^2}{x - 4} = \frac{-(x - 4)(4 + x)}{(4 - x)(4 + x)} \cdot \frac{1}{x - 4}$$

$$= \frac{-(4 + x)}{16x^2} = \frac{-4 - x}{16x^2}$$

$$f(4) = \frac{1}{4^2} = \boxed{\frac{1}{16}}$$

$$\lim_{x \rightarrow 4} \frac{-4 - (4)}{16(4)^2}$$

$$= \frac{-8}{16(16)} = -\frac{8}{256} = \boxed{-\frac{1}{32}}$$

10) Is the given function continuous? Justify your answer by drawing the graph.

$$f(x) = \begin{cases} x^2 & x \leq 0 \\ \sin x & 0 < x < \pi \\ \sqrt{x - \pi} & x \geq \pi \end{cases}$$

each piece is continuous on its own interval

we can check both 'joints' at $x=0$, $x=\pi$

(although, by inspecting the rules they look continuous)

$$\lim_{x \rightarrow 0^-} x^2 = (0)^2 = 0$$

$$\lim_{x \rightarrow \pi^-} \sin x = 0$$

$$\lim_{x \rightarrow 0^+} \sin x = 0$$

$$\lim_{x \rightarrow \pi^+} \sqrt{x - \pi} = \sqrt{0} = 0$$

$$f(0) = 0^2 = 0$$

$$f(\pi) = \sqrt{\pi - \pi} = \sqrt{0} = 0$$

Since left and right limit agree match the function value

continuous at $x=0$ and $x=\pi$

